

Galois Theory for Rats - Compass & Straightedge Construction

Skyler Hu

February 27, 2026

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1 Motivation & Construction Axioms

Initially, field theory and Galois Theory was all about solving ancient problems deemed unsolvable (in case of the quintic) so it is no surprise that it could be used to explain another collection of classic problems, that of compass and straightedge constructions. The notion of compass and straightedge constructions is usually introduced early on in math courses – we learned the basics of construction including constructing parallel lines, constructing the angle bisector of an angle and the perpendicular bisector of a line segment. There are nuanced differences between ‘construction’ problems we encountered in high school and ‘construction’ problems as accepted in field theory, so it serves us to first define what a compass and straightedge are and what can they do.

Definition 1.1. (Compass & Straightedge) A straightedge, unlike a ruler, has no measure of length. A compass is defined conventionally.

Definition 1.2. (Operations in Compass & Straightedge Construction) In compass and straightedge construction, one can only do the three following operations:

1. Draw a line through two given points;
2. Draw a circle centered at a given point and passing through another given point (i.e. given center and radius);
3. Obtain the point of intersection of a line and a circle or of two circles.

Note. A common mistake here is to assume that a compass *can preserve distance* – that is, one can copy a line segment of a given distance by aligning two ends of the compass to each end of the line segment. In the original (and classic) case of the construction problem posed by Euclid, a *collapsing* compass is used, one that would collapse when you pick it up. This is for two reasons: **(a)** Euclid made the fewest possible assumptions about the functions of a compass and straightedge, and **(b)** it is doubtful that the ancient Greeks had in their possession modern rigid compasses.

Despite this, it can be proved that the two compasses are equal in construction: what one could do, the other also could. A detailed explanation is included in the Proofs and Further Reading section.

The proofs of this section aren’t too difficult, so it is advised to work through them with a compass and straightedge if possible. The construction problems date back thousands of years and the resolution is extremely satisfying – have fun!

2 The Four Operations, plus one

When delving into compass and straightedge construction, we are most concerned about which lengths can be constructed from the unit length (1). The definition of constructibility helps us formalize this.

Definition 2.1. (Constructibility) A positive real number m is *constructible* if a line segment of length m can be constructed with compass and straightedge from a line segment of length 1, otherwise it is not constructible.

We might expect the compass and straightedge to be able to perform the four operations, addition, subtraction, multiplication and division. This is true, and we can prove it!

Theorem 2.1. (The Four Operations.) Given constructible lengths a and b , the lengths $a + b$, $a - b$, ab and a/b are all constructible.

Proof. We'll need the following elementary result:

Lemma 2.1. If a is constructible, then we can obtain on the plane the point $(a, 0)$ located on the x -axis. Generally, if a is constructible, then we can obtain a line segment of length a anywhere on the plane.

Note. We might consider this statement trivial, but in Euclid's original definition that uses a collapsing compass (non-distance preserving) this statement needs a rigorous proof. Details in the Proofs & Further Reading section. In the meantime, we can assume this is true.

The following proofs are geometric, so assume that a, b are positive and $a > b$ to avoid negative lengths.

1. Addition and Subtraction. This is easy. Construct a circle centered at $(a, 0)$ with radius b .

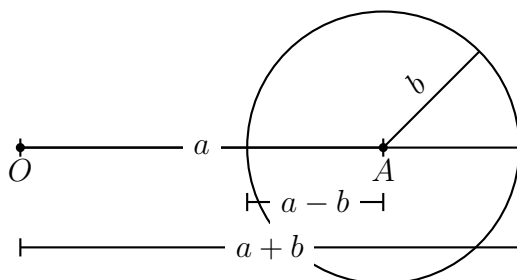
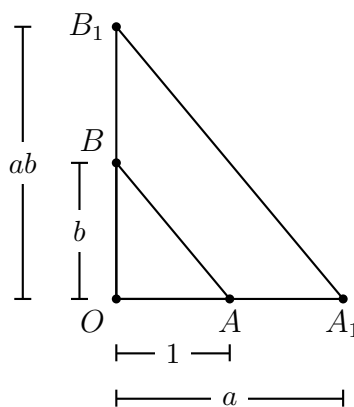


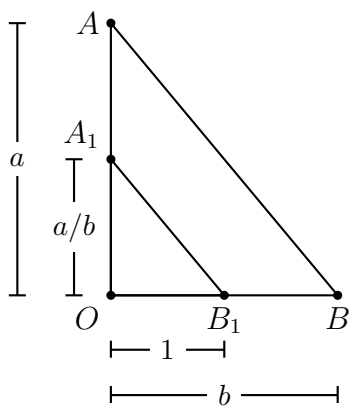
Figure 1: Addition & Subtraction

2. Multiplication. Consider constructing similar triangles.


 Figure 2: $\triangle OAB \sim \triangle OA_1B_1$

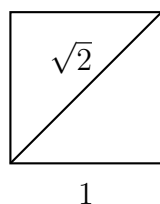
Construct from the origin O line segments $OA = 1$ and $OA_1 = a$. Construct a circle centered at O with radius b , pick an arbitrary point B on the circle, then $OB = b$. Construct line segment $A_1B_1 \parallel AB$ such that B_1 is on line OB . Then $OB_1 = ab$.

3. Division. Similar to the proof for multiplication, we construct two similar triangles with slightly different side lengths:


 Figure 3: $\triangle OAB \sim \triangle OA_1B_1$

Construct $OB_1 = 1$, $OA = a$, $OB = b$. Construct line segment $A_1B_1 \parallel AB$ with A_1 on line OA . Then $A_1B_1 = a/b$. $\square$

Therefore addition, subtraction, multiplication and division are all valid operations in compass and straightedge construction. We know that these aren't the only valid operations, though; take for example the fact that $\sqrt{2}$ is constructible.



$\sqrt{2}$ is the diagonal of a square of side length 1.

We can conclude that some roots are constructible. The constructible roots are none other than the square root, and we have the following:

Theorem 2.2. (Square root) If a is a positive constructible real number, then $\sqrt{a}$ is also constructible.

Proof. This one is slightly trickier and uses the elementary result that the angle subtended by the diameter is 90° (and also similar triangles). Construct line segments $AB = 1$, $AC = a$

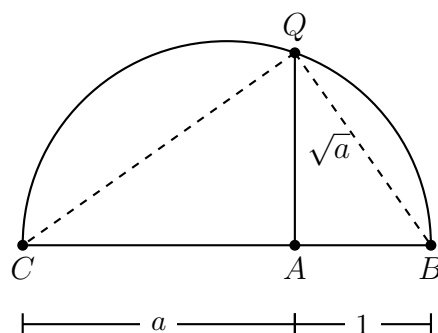


Figure 4: $AQ \perp BC$

with A, B, C collinear. Construct a semicircle with diameter BC , and construct $AQ \perp BC$ with Q on the semicircle. Then $\angle BQC = 90^\circ$, so $\triangle QAB \sim \triangle CQA$, and

$$\frac{AQ}{AB} = \frac{AC}{AQ},$$

so $AQ^2 = AB \cdot AC = a$. Therefore $AQ = \sqrt{a}$. $\square$

By now we have a solid grasp of what operations are valid in compass and straightedge construction: the four operations, along with taking square roots. In fact, we can show that these are the *only* valid operations, with an argument of (surprisingly) analytic geometry. We can formalize the ‘four operations & square root’ idea with field extensions:

Theorem 2.3. (Degree of Extension from Compass & Straightedge Construction.) If the element $\alpha \in F_k$ is obtained from field F through a series of compass and straightedge constructions, then $[F_k : F] = 2^k$ for some positive integer k .

Proof. We analyze each of the 3 defined operations (see **Defintion 1.2**).

1. Finding the intersection of two lines. A line in the plane with coordinates in F is of the form $Ax + By + C = 0$, with coefficients all in F . Solving for the intersection of two lines amounts to solving the simultaneous system of equations

$$A_1x + B_1y + C_1 = 0$$

$$A_2x + B_2y + C_2 = 0$$

which has solutions also in F .

2. Finding the intersection of a line and a circle. A circle in the plane is of the form $(x - h)^2 + (y - k)^2 = r^2$ with $h, k, r \in F$. Solving for the intersection of a circle and a line amounts to solving the simultaneous system of equations

$$Ax + By + C = 0 \tag{1}$$

$$(x - h)^2 + (y - k)^2 = r^2 \tag{2}$$

Using the substitution $x = -\frac{By+C}{A}$, we obtain from (2) a quadratic equation for y over F . Therefore the solutions x, y are in an extension of degree at most 2 over F .

3. Finding the intersection of two circles. This is

$$(x - h)^2 + (y - k)^2 = r^2$$

$$(x - h')^2 + (y - k')^2 = r'^2$$

Subtracting the two equations, we obtain a linear equation in terms of x and y (the quadratic terms cancel out). Then this case is the same as finding the intersection of a line and a circle, and the solutions x, y are in an extension of degree at most 2 over F .

We can then conclude that compass and straightedge construction gives an extension of degree 2^k over a base field F . $\square$

Note. This leads to the notion of the field of *constructible numbers*, including all real numbers in an extension of degree 2^k over the integers (since the base field of compass and straightedge construction is generated by 1).

3 The Classic Greek Construction Problems

Readers may have heard about (more or less) the three classic construction problems that have been troubling mathematicians since ancient times:

1. (Duplicating the Cube.) Given a cube, can we construct a cube with twice its volume?
2. (Trisecting the Angle.) Can we construct an angle that is $1/3$ of a given angle?
3. (Squaring the Circle.) Given a circle, can we construct a square with the same area?

It turns out that none of these three questions have an affirmative answer, as is often the case with longstanding suspicious unsolved problems (see: quintic, a big chunk of Gödel's work, parallel postulate) and we can disprove two of the three questions easily with what we've learned today! (You can take a guess at which problem is a lot harder than the other two.)

Theorem 3.1. (Classic Construction Problems.) None of the classic Greek construction problems (duplicating the cube, trisecting the angle, squaring the circle) is achievable through compass and straightedge construction.

Proof. 1. Duplicating the Cube. This problem is equivalent to asking whether or not $\sqrt[3]{2}$ is constructible, since a cube with volume 2 would have side length $\sqrt[3]{2}$. Since $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$ which is not an integer power of 2, $\sqrt[3]{2}$ is not constructible, and therefore it is impossible to obtain a cube with twice the volume of a given one with only compass and straightedge construction.

2. Trisecting the Angle. If an angle θ is constructible, then its cosine is constructible by finding the intersection of the angle and the unit circle. Our problem then becomes: given $\cos \theta$ is $\cos \theta/3$ constructible?

We can consider a special case (this is the most widely used argument for this problem): consider $\theta = 60^\circ$, $\cos 60^\circ = 1/2$. We only need to show that $\cos 20^\circ$ is not constructible. Using the triple-angle identity for cosine we obtain:

$$\cos \theta = 4 \cos^3 \theta/3 - 3 \cos \theta/3.$$

Substituting $\cos \theta/3 = x$ and $\cos 60^\circ = 1/2$, we have

$$4x^3 - 3x - 1/2 = 0 \Rightarrow (2x)^3 - 3(2x) - 1 = 0.$$

Let $y = 2x$. Then $y^3 - 3y - 1 = 0$, and by the Rational Root Theorem the only possible rational root of this equation is -1 , which is apparently not a root of $y^3 - 3y - 1$. Therefore $y^3 - 3y - 1$ is irreducible over $\mathbb{Q}$, and its roots are in an extension of degree 3 over $\mathbb{Q}$. Therefore y is not constructible and neither is $\cos 20^\circ$.

Note. The proof shows that it is not generally possible to trisect the angle. However, we can certainly trisect *some* angles, for example 180° and 360° (considering the regular hexagon which has an interior angle of 120° and an exterior angle of 60° , both 60° and 120° are constructible). It turns out that the only angles that are constructible are integer multiples of 3° , for more details see the Proofs and Further Reading section.

3. Squaring the Circle. A circle of radius 1 has area π , so this question is equivalent to asking whether or not π is constructible. The argument for classifying π as transcendental is well beyond the scope of this handout, but here we shall assume that $[\mathbb{Q}(\pi) : \mathbb{Q}]$ is infinite, so π is not constructible and it is impossible to construct a square with the area of a given circle. The proof of the transcendentality of π , unfortunately, is not included in the Proofs and Further Reading section because the author is also mystified. $\square$

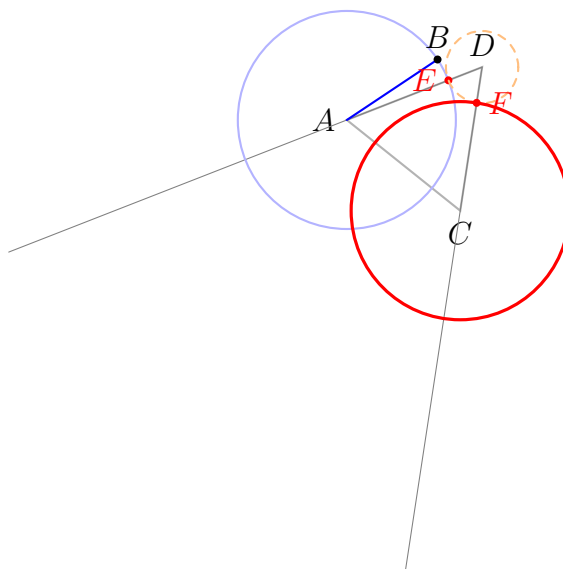
4 Proofs and Further Reading

4.1 More Tools

Euclid's initial definition of the compass and straightedge included a *compass* and a straightedge with no markings of distance. However, in the modern world, we encounter more frequently a *rigid* compass that can preserve the distance between two points, and a *ruler* which has markings of distance. How are the results different when we implement modern tools? First, we would assume the baseline: since a rigid compass and a ruler is 'better' than a collapsing compass and straightedge (at least in the conventional sense), we would expect them to be able to do anything a collapsing compass and straightedge could do. This is true, at least in the case of a rigid vs. collapsing compass:

Theorem 4.1. (Euclid's Proposition 2.) It is possible, using a collapsing compass and straightedge, to construct a line segment equal to a given line segment with one end at a given point.

Proof. Given line segment AB , we want to construct a line segment $CF = AB$ with endpoint at C .



1. Connect AC and construct equilateral triangle $\triangle ACD$.
2. Draw a circle centered at A with radius AB . Find the intersection of this circle and AD . Label it E .
3. Draw a circle centered at D with radius DE . Find the intersection of this circle and CD . Label it F . Then CF is a line segment with endpoint C and length $CF = AB$.

□

Euclid's Proposition 2 gives us that a rigid compass is equivalent to a collapsing compass: a rigid compass preserves distance, and a collapsing compass also could duplicate a distance (with some extra steps). What about a ruler? It turns out that a ruler is slightly more powerful than the straightedge (as one would expect it to be):

Theorem 4.2. (Ruler-and-Compass Construction.) If $a \in F'$ is obtained from a base field F using a series of ruler-and-compass constructions, then $[F' : F] = 2^m 3^n$ for some nonnegative integers m, n .

The ruler enables us to construct extensions of degrees 2 and 3 with its 'sliding' operation. Therefore, ruler-and-compass construction allows us to construct extensions of degrees 2, 3, 4 and their powers. What about degree 5? If you still remember what Galois Theory is about, you might have a very good answer for whether or not we can construct an extension of degree 5.

Corollary 4.1. Two of the 3 classic construction problems, (1) trisecting the angle and (2) doubling the cube, are obtainable using ruler and compass construction.

4.2 Less Tools

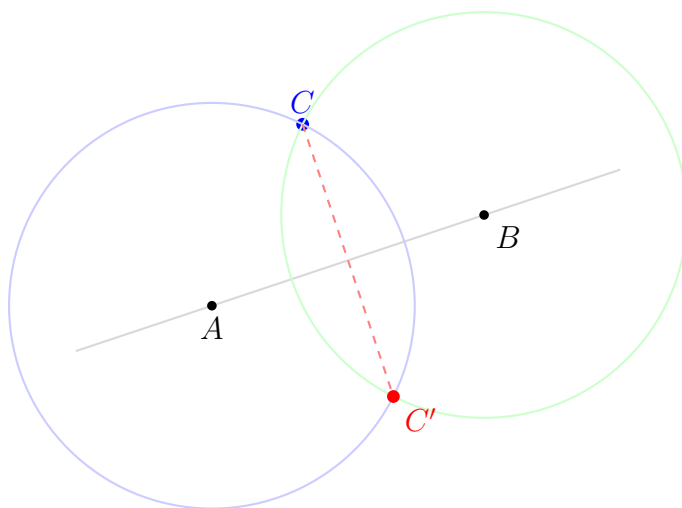
Let's consider the flip side, since mathematicians always like to go looking for trouble. Compass and straightedge construction requires *both* a compass and a straightedge. What happens if we only have a compass or a straightedge?

Compass-Only Construction. Suppose you don't like carrying around a straightedge - a piece of infinitely long wood can be unwieldy sometimes. Can you still do the construction problems as usual?

Theorem 4.3. (Mohr-Mascheroni Theorem.) Any construction that could be done with a compass and straightedge, can be done with a compass only.

Proof. We only need to show that the 3 basic operations can be done with compass only (see **Definition 1.2**). For length purposes (and the unfortunate discrepancy of literature on this specific subject), this handout only provides the 'building blocks' of the proof: (1) Reflection across a line, (2) Inversion through a circle, and (3) Finding the center of a circle through three points on the circle.

(1) **Reflecting a point across a line.** We can recall that compass and straightedge construction allows for reflecting points across a line. We can do this with compass only: Suppose we want to construct the point C symmetric across the line AB .

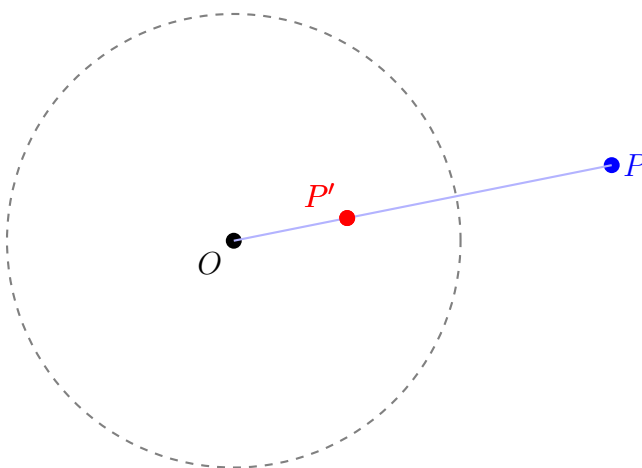


Construct two circles through C , one centered at A (blue) and one centered at B (green). There will be two intersections of $\odot A$ and $\odot B$: one is C , and the other is C' , the reflection of C across AB .

2. Inversion through a Circle. Before we start, we must be familiar with the definition of geometric inversion:

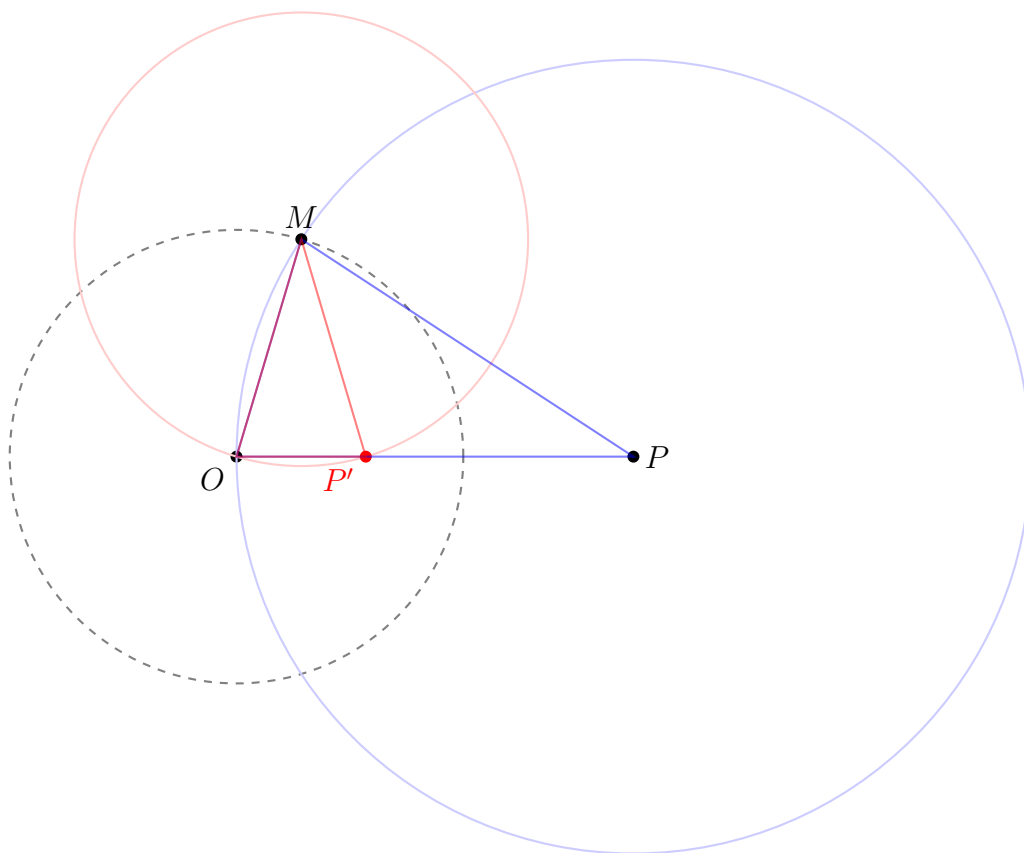
Definition 4.1. Inversion through a Circle. With respect to a reference circle $\odot O$ with radius r , the *inversion* of point P is the point P' lying on the ray OP such that

$$OP \cdot OP' = r^2.$$



In a sense, inversion with respect to a circle allows us to map points in a plane onto points in a circle, with points at infinite distance mapped to the center and points inside

the circle to points in the exterior. In the context of the Mohr-Mascheroni Theorem, we are more concerned about the former – how to map points of infinite distances on straight lines into a circle. You can visualize it as pulling a tissue through a ring by the center of the tissue: points at infinite distance fold and collapse into the circle. Here's how we do it with compass only (with similar triangles):



Suppose we want to find the inversion of P (in this case external to $\odot O$) with respect to $\odot O$.

1. Construct circle $\odot P$ with radius OP . There are two points of intersection of $\odot O$ and $\odot P$: they are symmetric, so pick one of them and label it M_1 . The other one is M_2 (not shown).
2. Construct circle $\odot M_1$ with radius OM_1 (i.e. radius of $\odot O$) and find its intersection with OP . Label it P' . Then P' is the inverse of the point P relative to $\odot O$.
(If you're wondering how to locate P' when we don't have the ray OP : OP is the perpendicular bisector of the line segment M_1M_2 , so we can find P' by drawing two arcs centered at M_1 and M_2 both with radius $= OM$.)

We can show that P' is the inverse: since $\triangle MOP$ and $\triangle MOP'$ are both isocles with a common bottom angle, $\triangle MOP' \sim \triangle PMO$ and

$$\frac{OP'}{OM} = \frac{OM}{OP}.$$

Therefore $OP' \cdot OP = r^2$ where r is radius of $\odot O$.

3. Finding center of circle through 3 given points. This problem is known as one of *Napoleon's Problems*, consisting of a couple compass-only construction problems. Napoleon, an amateur mathematician himself, had reputedly been related to this problem. (When you're French, mathematical and revolutionary...)

□

Straightedge-Only Construction. Suppose your compass was confiscated at the TSA for being a public safety hazard. In that case, you might want this:

Theorem 4.4. (Poncelet-Steiner Theorem.) Any construction that could be done with a compass and straightedge, can be done with a straightedge only, provided that there is already a circle on the plane.

Nothing – only paper. Suppose you left your compass and straightedge at home but still came to algebra class anyway. You pick up the paper you were supposed to draw diagrams on remorsefully. Then you turn it over in your hand, and an idea comes to mind. Can you perform the classic constructions using nothing but folding the paper?

With origami, folding paper as a means of construction, we enter into the realm of *Huzita-Hatori Axioms*, a set of axioms that dictate what can be done with paper folding just as the three rules to compass-straightedge construction. The full set of axioms is given here:

Theorem 4.5. (Huzita-Hatori Axioms.) The 7 following constructions are allowed using origami:

1. Given two distinct points P_1 and P_2 , there is a unique fold passing through both of them;
2. Given two distinct points P_1 and P_2 , there is a unique fold that places P_1 onto P_2 ;
3. Given two lines l_1 and l_2 , there is a fold that places l_1 onto l_2 ;
4. Given a point P_1 and a line l_1 , there is a unique fold perpendicular to l_1 that passes through point P_1 .
5. Given two points P_1 and P_2 and a line l_1 , there is a fold that places P_1 onto l_1 and passes through P_2 .
6. Given two points P_1 and P_2 and two lines l_1 and l_2 , there is a fold that places P_1 onto l_1 and P_2 onto l_2 .
7. Given one point P and two lines l_1 and l_2 , there is a fold that places P onto l_1 and is perpendicular to l_2 .

You can grab a piece of paper and try out these constructions. Even though you've probably encountered origami in kindergarten, remember that for whatever thing you think of there is a group of people (usually mathematicians) who hold it to the highest interest!

4.3 What's Next?

This handout – on compass and straightedge constructions – has begun as a required course for anyone into field theory. Construction is an extremely rich topic and there are still many things we haven't covered, including:

- Full proofs of the Mohr-Mascheroni, Poncelet-Steiner theorems and the Huzita-Hatori axioms;
- Euler's construction of n -gons where n is a Fermat prime, specifically the construction of the 17-gon (and I'm terrifically sorry for this exclusion – I know it's a fan favorite and has very important implications for field extensions!);
- Euclid's Propositions (the things we take for granted);

these provide a guide for anyone fascinated by construction problems and hoping to explore further.